Innovation & Entrepreneurship for Engineers

Dr Shankar

IIT Madras

2026

Innovation & Entrepreneurship for Engineers

Module 1
Engineers as Innovators

#1. What is Innovation ?
#2. Innovation through S & T

Module 2
Problem Discovery

#3. Finding Problems worth solving
#4. Customer Discovery & Design Thinking

Module 3
Creativity & Innovation

#5.Innovative Thinking – TRIZ, FLOW, CREATE
#6. Enhancing Innovation – AI - Collaboration

Module 4
Deep-Tech Entrepreneurship

#7. Deep – Tech Ventures
#8. Exponential & Systems Thinking

Module 5
Building a Startup

#9. Economics of Innovation & Entrepreneurship
#10. Business Models
#11. Intellectual Property

Module 6
Prototyping & Presentation

#12. Rapid Prototyping
#13. Pitch Your Idea
#14. Demo Day

Innovation Equation

Innovation = Idea + New Value + Path to Market

Innovation is the specific instrument of entrepreneurship.
It is the act that endows resources with a new capacity to create wealth – Peter Drucker

Idea / Invention / Innovation – Electric Bulb

The Idea

Edison thinking,

"We should find a way to use electricity to make safe indoor light to replace gas lamps"

Incandescence - Known Science – New Application

The Invention

Edison and his team at Menlo Park testing thousands of filaments—such as carbonized bamboo—to make a bulb that actually glowed and stayed lit.

Electric Bulb - Prototype – Patent – Team Effort

The Innovation

Edison building the Pearl Street Station power plant, laying cables under streets, and selling electricity so people could actually flip a switch and use the light bulb in their daily lives.

Electric Illumination – scaleup - commercialization

From Idea to Innovation

Idea → **Idea - Problem** → **Desirable Idea**
Scalable Idea
Sustainable Idea → **Feasible Idea**
Patentable Idea → **Viable Idea that creates value to the Customer**



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Innovation through Science & Technology

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Objective of Session 2

- Many breakthrough innovations originate from fundamental scientific discoveries.
- This session explores how advances in physics, chemistry, biology, mathematics and computer science have led to transformative technologies such as semiconductors, GPS, lithium ion batteries, AI, Aadhaar and UPI.
- Students are introduced to *Pasteur's Quadrant*, which highlights how the most impactful innovations often emerge from research that advances scientific knowledge while solving important real-world problems.
- Framework: Science – Technology – Product – Startup - Industry

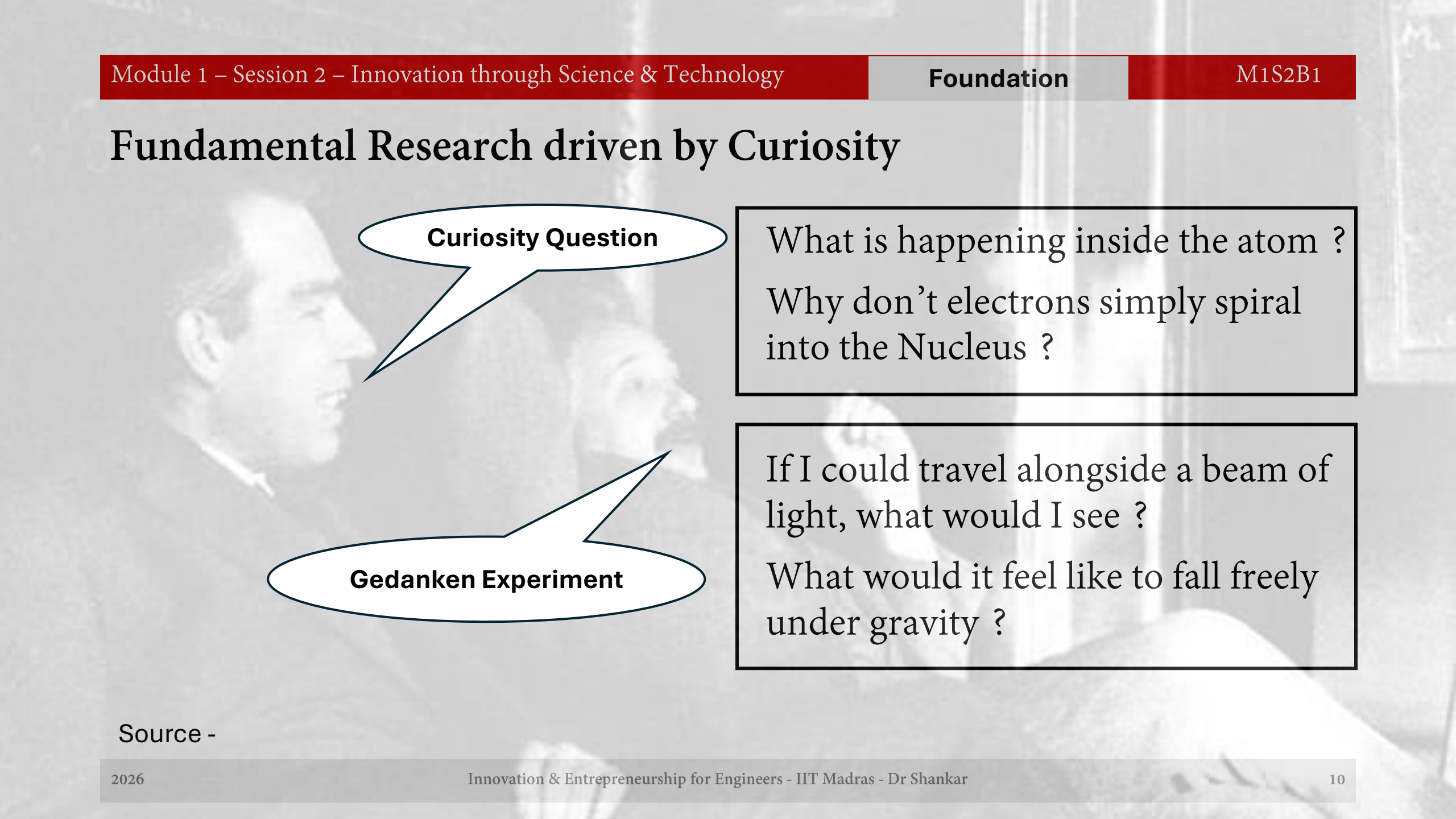
Innovation through Science & Technology

- Foundation – Key Concepts
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- Practice – Examples and Case Study
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Innovation – Foundational Concepts

- Curiosity driven Research
- Application driven Research
- The Pasteur's Quadrant – where Curiosity meets Application
- Summary of Key Ideas

Fundamental Research driven by Curiosity



Curiosity Question

What is happening inside the atom ?

Why don't electrons simply spiral into the Nucleus ?

Gedanken Experiment

If I could travel alongside a beam of light, what would I see ?

What would it feel like to fall freely under gravity ?

Source -

Which of these questions has an obvious commercial application ?

1. Why don't electrons collapse into the nucleus ?
2. Why does light always appear to travel at the same speed ?
3. What would I see if I travelled alongside a beam of light ?
4. Why do atoms emit only specific colors ?


CAUTION
CUTTING




QBIT FORCE


Tunnel Diode



The fundamental scientific discoveries that enable GPS

GPS determines where you are
by measuring how long precisely timed radio signals
take to travel from satellites to your receiver.

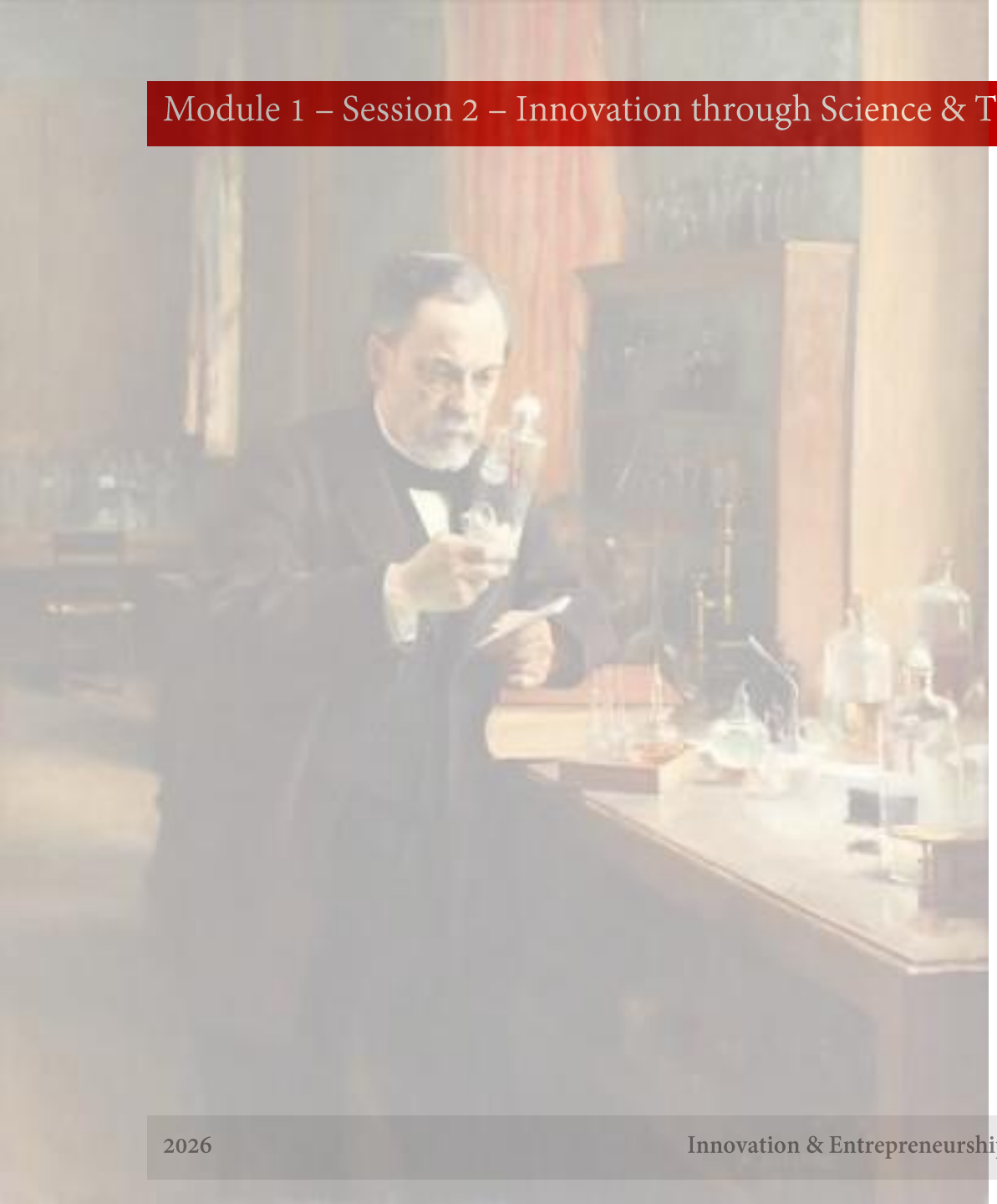
- Electromagnetism – Maxwell
- Speed of Light – Maxwell + Einstein
- Special Relativity – Einstein
- General relativity – Einstein
- Atomic Physics – Quantum Mechanics (1 microsecond – 300m)
- Geometry – Euclid…
- Statistical Estimation & Information theory – Signal Vs Noise

Application – driven Research

- Can we create an electric light that is bright, safe, affordable and practical enough to replace the gas lamp ?
  - What material can survive at high temperatures ?
  - How can the filaments be designed to last hundreds of hours ?
  - How to manufacture thousands of identical bulbs ?
  - How do we distribute electricity?
  - How do we switch the lights on and off ?

What must I invent to make this technology useful to ordinary people ?

Source -

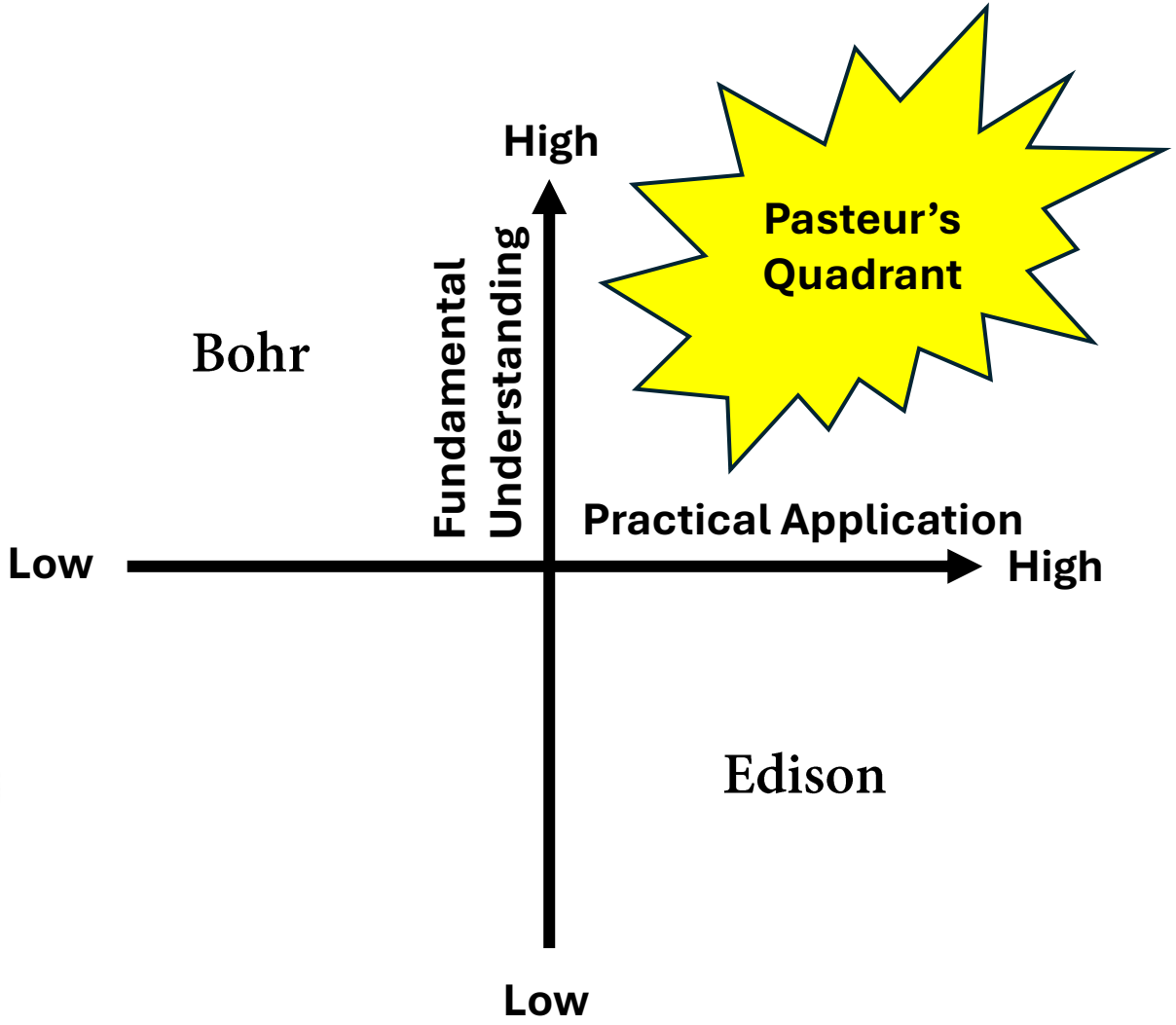


Louis Pasteur had a different approach !

- Start with a problem that matters – fermentation of wine, sick silkworms, spontaneous generation...
- Go deep into the science to discover something fundamental
- Then translate that knowledge into something that changes the world.
- If a microorganism can cause disease, can we somehow use a weakened form of it to teach the body how to fight the real disease ?

Pasteur's Quadrant





Source -

Pasteur's Questions

- Why does this happen – Scientific Curiosity
- What fundamental mechanism is responsible – Deep Understanding
- Can that understanding solve an important problem ?
- Can the solution be translated into practice ?

Why does this happen – and can understanding it help us to solve an Important problem ?

Source -

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Dr Premnath Venugopalan

- Dr Premnath is Founder Director, Venture Center (National award winning inventive enterprises and deep tech incubator) and a scientist at CSIR-National Chemical Laboratory, Pune, India.
- Dr Premnath is a technology developer, innovation and incubation manager, startup mentor and a co-founder of 2 medtech startups. One of his inventions -- a breakthrough material for hip and knee joint replacements — has been implanted in more than a million patients worldwide.
- Another technology for porous maxillo-facial implants has been implanted in thousands of patients in India and abroad. He has provided leadership for teams that have won National awards for technology development of medical products, intellectual property management and business incubation.



He is chemical engineer and an alumnus of MIT in the US, IIT-Bombay (Distinguished Alumnus, 2022) and has been a Chevening Technology Enterprise Scholar in Cambridge, UK.

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Innovation Canvas for Case Study Analysis

- Problem – What Customer Problem was solved ?
- Technology – What Scientific and Engineering advances enabled the Solution ?
- Customer Value – Why would Customers buy it ?
- Business Model – How does the Company Create and Capture Value ?
- Ecosystem – What complementary infrastructure or partners were needed ?
- Timing – Was the Market ready ?
- Key Risks – What Assumptions could have gone Wrong ?

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Innovation through Science & Technology – Key Ideas

- Innovations through Science & Technology
- Curiosity – driven research – Fundamental discoveries – Bohr's Quadrant
- Application – driven research – Edison's Quadrant
- Curiosity meets Application – Pasteur's Quadrant
- Dr Premnath's Case Studies of Indian Deep Science / Tech Startups

Innovation through Science & Technology – Assignment M1S2B1

Choose of one of the deep tech startups – Ahammune / KB Cols / Flytbase (Drones) – study their case and analyse using the Innovation Canvas (Chart #22)

For details of the first two startups refer to Dr Premnath's presentation deck (PDF) and for the third, conduct a search and gather relevant details

This is a group assignment – desired format is PPT - it is due by 26th August 9 pm

Format –

Innovation through Science & Technology – References

- https://www.science.org/doi/10.1126/science.adx3736
- https://bigthink.com/business/how-pasteurs-quadrant-enlightens-the-invention-innovation-challenge/
- https://www.catalign.in/2008/10/bangalore-innovation-forum-and-pasteurs.html